

ISX-3 / ISX-3mini Datasheet

Sciospec Scientific Instruments GmbH - ISX-3 / ISX-3mini



Accessories

What is included

- ISX-3 / ISX-3mini
- power cord or external power supply
- full speed USB cable
- USB flash drive with software and manual

Optional accessories

Kelvin clips

(PN: SA000275)



component test
fixture 10 MHz

(PN: EA002764)



component test
fixture 30 MHz

(PN: EA002793)



SMD test fixture
(PN: EA002733)



Kelvin clips
fixture
(PN: EA002735)



dielectric test
fixture
(PN: EA002763)



liquid test fixture
(PN: EA002792)



compensation kit
(PN: SA000278)



Key Features

- one-channel compact impedance analyzer
- second multiplexed channel (ISX-3mini default, ISX-3 optional)
- 2, 3, and 4-port measurements
- single sine excitation at a wide frequency range of 100 mHz ... 10 MHz (optional: 40 MHz or 100 MHz)
- up to 0.01% base accuracy
- measurement speed up 0.5 ms/point
- compatibility to test adapters for electrical components, foils, thin films, and liquids
- removal of parasitics via open-short-load compensation
- intuitive software with data visualization in spectral and time trace view
- full speed USB interface
- SCPI interface for external communication
- included LCR View
- Optional AddOns like Multiplexers and Battery available

Typical Fields of Applications

- quality inspection and production
- research, development, and prototyping
- maintenance, repairing, and troubleshooting
- calibration, metrology, and education
- special applications in industry and engineering (high-frequency engineering, medical engineering, automotive, energy engineering, etc.)

Technical Data

Measurement parameters		
Frequency of test signal	Frequency range	100 mHz ... 10 MHz (optional: 40 MHz or 100 MHz)
	Frequency resolution	$f < 10 \text{ kHz}$: $< 6 \text{ mHz}$ $10 \text{ kHz} \leq f \leq 100 \text{ kHz}$: $< 24 \text{ mHz}$ $f > 100 \text{ kHz}$: $< 117 \text{ mHz}$
	Frequency precision	$\pm 100 \text{ ppm}$ (@ 25°C) - additional max. $\pm 10 \text{ ppm}$ over specified temperature range (e.g., at 0°C) - additional $\pm 5 \text{ ppm}$ during first year
Excitation signal (C port)	Voltage range of excitation signal	10 mV ... 1 V (peak amplitude)
	Resolution of voltage signal	0.1 mV
	Precision of voltage signal (uncalibrated)	$\pm 1\%$ typ. ($\pm 8\%$ max.)
	Current range of excitation signal	10 μA ... 10 mA (peak amplitude)
	Resolution of current signal	1 μA
	Precision of current signal (uncalibrated)	$\pm 1\%$ typ. ($\pm 8\%$ max.)
Potential measurement (R & WS ports)	Measurement range	$\pm 1 \text{ V}$
	Resolution of measurement range	$< 0.008\%$
	Precision of measured potential	- uncalibrated gain error 1% max. - uncalibrated offset error 2% max. - electrometer amplifier input bias current: $\pm 1.5 \text{ pA}$ typ. ($\pm 50 \text{ pA}$ max.)
Current measurement (W port)	Measurement ranges	$\pm 10 \text{ mA}$ $\pm 100 \mu\text{A}$ $\pm 1 \mu\text{A}$ $\pm 10 \text{ nA}$
	Resolution of measurement range	$< 0.008\%$
	Precision of measured current	- uncalibrated gain error 0.1% max. for $\pm 10 \text{ mA}$ range 0.1% max. for $\pm 100 \mu\text{A}$ range 1% max. for $\pm 1 \mu\text{A}$ range 5% max. for $\pm 10 \text{ nA}$ range - uncalibrated offset error 2% max. - current amplifier input bias current: $\pm 10 \text{ pA}$ typ. ($\pm 1 \text{ nA}$ max.)

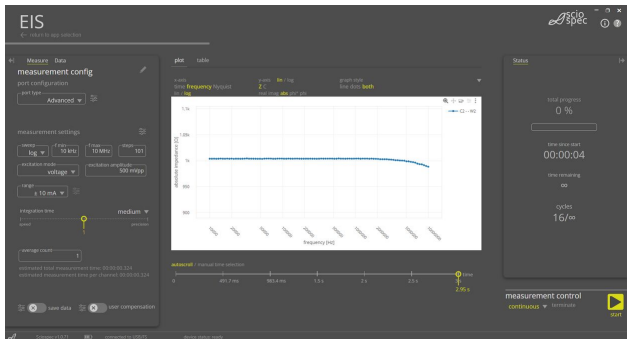
Measurement parameters (continued)	
Output impedance	20 Ω
Basic accuracy	
Sciospec LCR calculated accuracy contour plot	
Measurement speed	
Ranges of equivalent circuit models	
$ Z , Z_{\text{real}}, R, R_p, R_s$	1 m Ω ... 1 T Ω
Z_{imag}	$\pm 1 \text{ m}\Omega$... $\pm 1 \text{ T}\Omega$
$ Y , G$	1 pS ... 1 kS
B	$\pm 1 \text{ pS}$... $\pm 1 \text{ kS}$
θ	-90° ... 90°
C, C_p, C_s	10 fF ... 1 kF
L, L_p, L_s	1 nH ... 1 TH

General specifications	
Power requirements ISX-3mini	12 V DC (typ.), 15 W (max.), connector DC jack (type: Switchcraft PN: 712A)
Power requirements ISX-3	100-240V AC (typ.), 50/60Hz, 60W (max)
Timing accuracy	1% accuracy over specified temperature range
Operating conditions	- temperature range: 0 °C ... 40 °C - relative humidity: ≤ 80% (non-condensing) - altitude: 0 m ... 3000 m
Non-operating conditions (i.e., storage conditions when the device is turned off)	- temperature range: -25 °C ... 80 °C - relative humidity: ≤ 80% (non-condensing) - The temperature gradient should not exceed 1 K/min to reach operating conditions.

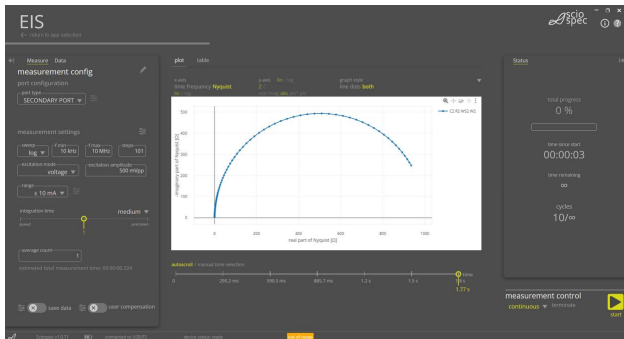
General specifications (continued)	
Dimensions ISX-3	248.67mm x 97.1mmx 193.2mm (width x height x depth)
Weight ISX-3	2.5 kg (typ.) depending on options
Dimensions ISX-3mini	112 mm x 58 mm x 212 mm (width x height x depth)
Weight ISX-3mini	1.0 kg (typ.)
Master interface	isolated full speed USB interface - standard conformity: USB 2.0, FS (12 Mbit/s) - connector: Mini USB 2.0 - ESD protection: class 3A contact ESD performance per ANSI/ESD STM5.1-2007 Ethernet interface - standard conformity protocol10/100 Base-T, RJ45 - protocol TCP/IP
Connector type ISX-3	4 BNC connectors (22 mm separation) for 2-, 3-, and 4-point configurations (other connectors, e.g. SMA, MCX, on request)
Connector type ISX-3mini	2 times 4 MCX connectors for two multiplexed 2-, 3-, and 4-point configurations

Software Overview

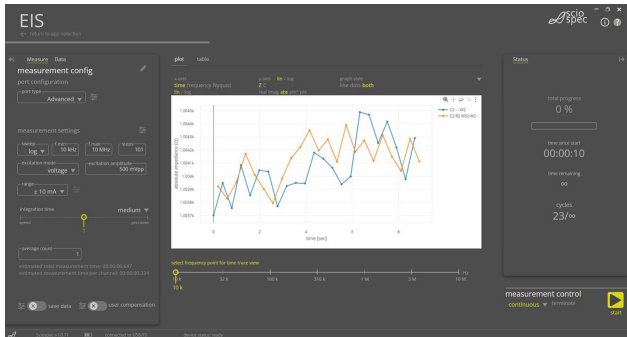
Z EIS view



Nyquist Plot



Time trace view



LCR View

